MINOR ASSIGNMENT

Web Application Projects With React

(CSE 3395)

Submitted by

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Abstract

The Web Scraper MERN Application is an interactive, full-stack software system built with MongoDB, Express.js, React.js, and Node.js. It allows users to submit website URLs, extracts structured DOM components including hierarchical headings (H1–H3), navigational hyperlinks, and e-commerce product catalogs, and preserves records within a cloud MongoDB database. Deployed live on Vercel and Render, the system provides an intuitive interface with automated input validation and high reliability.

Introduction

Automated web extraction is fundamental in web engineering for data harvesting, competitive analysis, and search engine indexing. This project implements a decoupled Model-View-Controller architecture featuring an asynchronous React Single Page Application communicating via RESTful API endpoints with an Express.js backend. Cheerio is utilized for document parsing, Axios for HTTP transfers, and Mongoose ORM for structured object-data modeling with MongoDB.

Frontend Components:

The client interface is developed using React 18, Vite, and Lucide icons, organized into four primary modular components: (a) **Navbar (Navbar.jsx)**: Provides persistent application branding, tabbed navigation, and dynamic MongoDB connection status monitoring; (b) **HomePage (HomePage.jsx)**: Features the interactive scraping console with input URL validation, quick preset website chips, and submission controls; (c) **ResultsPage (ResultsPage.jsx)**: Renders extracted web content across tabbed views for Headings Hierarchy, Hyperlinks, and Product cards; (d) **HistoryPage (HistoryPage.jsx)**: Displays an interactive historical table with timestamps, record inspection, and deletion modals.

Database :

The persistence tier uses a cloud MongoDB Atlas cluster (database: *webscraper*, collection: *scraped_data*) managed via Mongoose schemas. The schema defines structured attributes: *url* (String), *pageTitle* (String), *scrapedDate* (Date), *headings* (Array of level and text), *links* (Array of label and href), and *products* (Array of product details). The system supports full CRUD operations with indexed queries and fallback in-memory caching during connectivity transitions.

Screenshot as Outputs

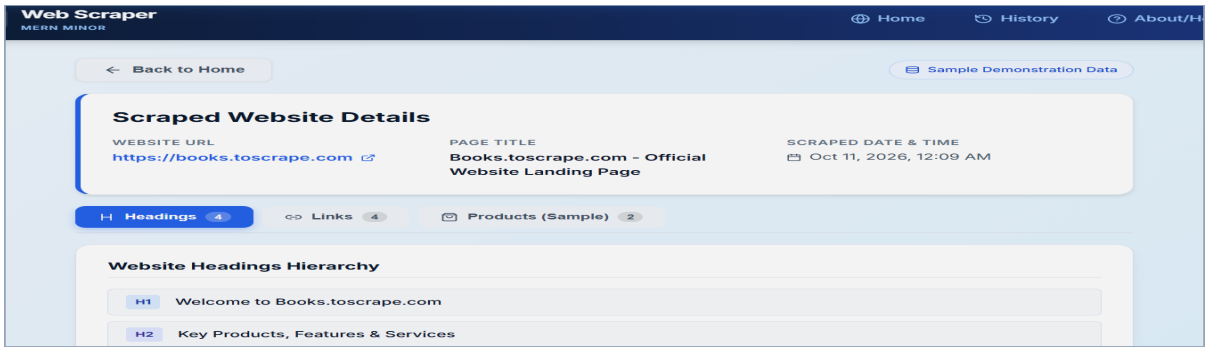


Figure 1: Live Web Scraper Output showing extracted headings hierarchy, metadata, and active MongoDB cloud status.

Conclusion

The Web Scraper MERN Application fulfills all functional and academic requirements for the CSE 3395 minor project. It demonstrates full-stack integration, robust asynchronous REST communication, structured DOM parsing, and cloud database persistence, successfully deployed on Vercel and Render.